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Forecasting Crude Oil Prices: Evidence From WOA–VMD–FE–Transformer Model

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optimized WOA-VMD method is used to decompose the crude oil futures price series into multiple sub-sequences. Then, according to the fuzzy entropy (FE) value of the sub-sequence, it is reorganized into three subsequences: low frequency, medium frequency and high frequency. Finally, we trained these three subsequences using the Transformer model and applied them to the test set to make predictions. By adding the forecasts together, we get the final forecast and use multiple metrics to assess the accuracy of the forecast. After experimental verification, WOA-VMD-FE-Transformer model shows high accuracy in predicting crude oil futures prices. The application of the model presented in this study can help decision-makers develop strategies, help investors make informed investment decisions, and also promote the equilibrium between energy supply and demand in the market. Hence, the research holds considerable importance in fostering the sustainable growth of the energy market and related industries.

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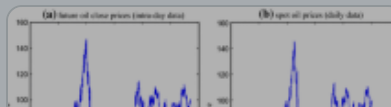
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